Riley Beenders

Lead, R&D Engineer

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RileyBeenders - MyFolio Project

PROFILE

Lead R&D Engineer with prior CTO experience driving product development, manufacturing, and industrial automation for complex electromechanical systems. Expertise in DFM, precision tooling and fixture design, CAD, production workflow optimization, and cross-functional mechanical, electrical, firmware, and software integration. Proven ability to take advanced prototypes through validation, assembly, and commercialization.

EXPERIENCE

Proteor / Proteor Printing Solutions

Lead, Research & Development Engineer — *Coplay, PA* · **Jan 2025 - Present**

Proteor acquired Filament Innovations in January, 2025 and my role evolved from CTO to Lead R&D Engineer. I continue to carry out my former CTO responsibilities, now with expanded oversight of the East Coast engineering team and involvement across Proteor's global product portfolio.

- Led the design and development of the ICARUS-Lite, a compact 3D printer for digital prosthetic and orthotic production, including manufacturing process development for custom AC heaters with dual-voltage 120/240 VAC capabilities. The dual-voltage heater implementation increased internal production by 30%.
- Led the ICARUS-Lite from product development through commercialization, applying design for assembly (DFA) and production workflow optimization to support increased revenue and the addition of approximately five new roles.
- Developed an automated LabVIEW system that collects nine data points every 15 ms from multiple laser sensors over Modbus, charts tolerance data, and supports geometric dimensioning and tolerancing (GD&T) analysis for internal quality control and end-customer use.
- Led the implementation of Microsoft Business Central across the division's product portfolio, integrating data with global Proteor locations to streamline sales and inventory management.
- Led the implementation of ProteorPrint OS, a network-accessible remote operation and diagnostics interface with live video, AI-assisted alerts, and material-waste metrics, reducing equipment downtime and enhancing remote support.
- Performed and documented risk analysis covering safety, operational, and software risks for ICARUS and ICARUS-Lite products along with the ProteorPrint cloud platform, implementing mitigations applicable to EU and U.S. standards.
- Collaborated with another additive manufacturer to develop and validate a reliable, cost-effective hotend that met the material's specific needs.
- Worked alongside production, sales, and customer-service teams to support the product line, providing hardware, electronics, motion-control, software troubleshooting, and after-sales technical support, which increased customer-satisfaction.
- Converted the Filament Innovation POSEIDON and ICARUS platforms from servo-driven ballscrews to a standardized linear-motor architecture, achieving 1-micron precision per 1 m/s while moving a 30 kg payload. The system reached a maximum speed of 4 m/s.
- Continued development of our internal IT infrastructure and standardized software and internal processes within our division of Proteor

Filament Innovations

Chief Technology Officer — *Whitehall, PA* · **Jan 2023 - Jan 2025**

Acquired by Proteor and is recognized globally as 'Proteor Print'.

- Oversaw the product-development pipeline, internal IT infrastructure, and technology strategy while retaining hands-on responsibility for mechanical, electrical, firmware, and software systems.
- Led the design and launch of the Gen3 POSEIDON, applying fixture design and design for assembly (DFA) principles to a system combining a 1 m^3 build volume, filament-and-pellet dual extrusion, braked NEMA 34 Z drives, industrial servo XY motion, and a custom gantry.
- Designed and hosted a redundant Kubernetes cluster for a self-developed ERP/CRM that managed orders, inventory, tasks, projects, and customer interactions, with dynamic BOM generation linked to purchasing, which cut order-processing time by almost 65% and increased inventory accuracy.
- Researched and developed high-speed linear motor systems that moved a 30 kg payload at 3 m/s with 3-micron accuracy, validating test results against performance targets. Worked with overseas suppliers to coordinate manufacturing and support deployment in upcoming ProteorPrint models.

Lead Electro-Mechanical Engineer — *Whitehall, PA* · **Jun 2022 - Jan 2023**

- Improved thermal and acoustic performance through enclosure redesign and component placement, including custom tooling with local fabricators for large-radius aluminum-panel bends.

- Integrated a multi-board CAN-bus control architecture that simplified wiring and debugging while supporting larger, modular machines.
- Developed independently testable electronics enclosures that could be installed in minutes and coordinated custom harnesses, panel designs, and hardware improvements with vendors.

Industrial Engineer *Whitehall, PA* · **Apr 2021 - Jun 2022**

- Designed and released KRATOS, a machine that prints high-strength above and below-knee prosthetic sockets faster.
- Developed next-generation ICARUS and POSEIDON models with higher-performance components and expanded motion safety through NEMA 34 motors, planetary gearboxes, and power-failure brakes.
- Designed THE KRAKEN, a 1 x 2 x 1.5 m hybrid pellet-and-filament gantry printer for a client producing the world's largest 3D-printed statue, and managed CAD, BOM, sourcing, fabrication, assembly, firmware tuning, support, and training.

Engineering Technician *Whitehall, PA* · **May 2020 - Apr 2021**

- Diagnosed mechanical and electrical problems during assembly and operation, improving early product reliability.
- Tuned, calibrated, and optimized print parameters while documenting design limitations and opportunities for product improvement.

RJB.Engineering

Owner & Engineering Consultant *Pennsylvania* · **2015 - Present**

- Deliver project-based engineering support for automotive clients utilizing CATIA and MATLAB.
- Produced 3D-printed plastic face shields using a custom designed printer, supplying local healthcare providers during the COVID-19 pandemic to meet urgent protective-equipment needs.

Blue Mountain Ski Resort

Ski Lift Operator *Palmerton, PA* · **Nov 2019 - Jan 2021**

- Operated lifts and assisted thousands of guests with ticketing, weather, and slope information.
- Performed motor and control safety checks and supported the maintenance team during lift breakdowns.

Northampton Area School District

Computer / Network Technician *Northampton, PA* · **Jun 2016 - Aug 2019**

- Assisted with my school district's network infrastructure, installing over 750 wireless access points across six school buildings, which increased Wi-Fi coverage and reduced connectivity issues.
- Accelerated deployment of more than 2,500 Chromebooks by developing an Arduino-based tool to automate setup-data entry.
- Managed the final distribution of devices with school administrators, creating detailed documentation and checklists that ensured all students received the correct equipment.

EDUCATION

Penn State University **Dec 2024**

BS, Electro-Mechanical Engineering

CERTIFICATIONS

- MATLAB OnRamp: MathWorks | Issued July 2026
- Simulink OnRamp: MathWorks | Issued July 2026
- Simscape OnRamp: MathWorks | Issued July 2026
- Machine Learning Onramp: MathWorks | Issued July 2026
- Lean Six Sigma Yellow Belt: Educate 360 | Issued July 2025
- Lean Six Sigma White Belt: Educate 360 | Issued June 2025
- Certificate in Engineering Design and Tools: Penn State University | Issued Spring 2023
- Mechanical Engineering Certificate: Lafayette College | Issued Spring 2019

SKILLS

- **Engineering**: Design for Manufacturing, Design For Assembly (DFA), Engineering Documentation and BOM Control, Manufacturing Process Development, Product Development and Commercialization, Product Safety and Regulatory Compliance, Production Workflow Optimization, R&D Prototyping & Validation, Technical Writing and Documentation-, Vendor Communication and Cost Negotiation, Worldwide Project Collaboration, Lean Manufacturing, Additive Manufacturing, Work Instructions And SOP Writing, Geometric Dimensioning & Tolerancing (GD&T)
- **Design & Software**: Autodesk Inventor, CATIA, Fixture and Tooling Design, Fusion 360, GitHub, LabVIEW, LM Studio, Macros, MATLAB, oMLX, Python, Rapid Prototyping, Siemens NX, SolidWorks, Visual Studio Code
- **Controls & Automation**: Allen Bradley PLCs, Data Logging and Process Analysis, Git Version Control, Workflow Automation, Automation Equipment Development, Jig Design
- **Infrastructure & Systems**: Domain-Based Application Access, DNS and Subdomain Configuration, Docker and ContainerizedApplications, Proxmox Virtualization, Reverse Proxy and Firewall Routing, Self-Hosted Infrastructure, Server Network and VLAN Configuration, Virtual Machine and Container Management, Windows/Linux Systems Administration